

Course	Embedded and Mobile Applications
Document	AME26_27_OPTIONAL_exercise2.pdf
Date	02/10/2026
Description	Optional exercise for creating library for the photoresistor application
Scope	Optional.

Statement.

You must create an application that controls the public lights of a city. In order to do it, you need to use a Photoresistor, that basically sense the ambient light, when the ambient light sensed is below to a threshold you need to open the city lights, and when the ambient light is up to a threshold you need to close the city lights.

Also, you need to add the possibility to modify the threshold using the rotary angle sensor, and as there are two possible thresholds you can control the threshold to modify pressing a button.

As you know you can see, we need to use hysteresis.

Hysteresis

Technique used in electronic and embedded systems to prevent undesired oscillations caused by noise or small fluctuations in an input signal. Instead of using a single threshold to change the state of a system, hysteresis employs **two distinct thresholds**:

- An **upper threshold** that triggers a transition from one state to another.
- A **lower threshold** that triggers the reverse transition.

The region between these two thresholds is known as the **hysteresis band**. Within this band, the system maintains its current state.

Principle of Operation

Consider a light-activated lamp. Without hysteresis, the lamp might turn ON when the measured light level falls below a threshold of 50%, and turn OFF when it rises above the same threshold. If the sensor reading fluctuates around 50% due to measurement noise, the lamp may repeatedly switch ON and OFF.

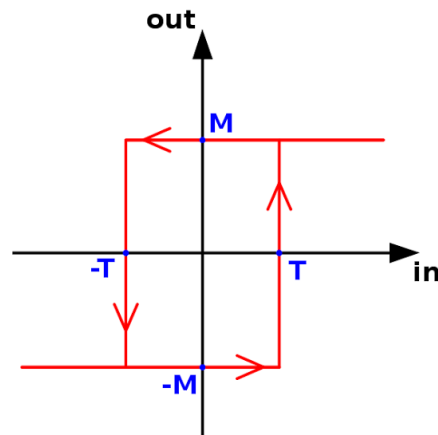
With hysteresis, two thresholds are defined:

- Lamp turns **ON** when the light level falls below 40%.
- Lamp turns **OFF** when the light level rises above 60%.

As a result:

- If the lamp is OFF, it remains OFF until the measured value drops below 40%.
- Once the lamp is ON, it remains ON until the measured value exceeds 60%.

This separation between switching points eliminates rapid state changes and improves system stability.



Simulation

To simulate the city lights you can use a LED and to show the threshold limits you can use the debug console.

Connecting the board

To make this exercise we need to use the following:

- Photoresistor sensor.
- Rotary angle sensor.
- Button
- LED actuator
- Led
- Wires
- Nucleo board + Shield to connect the sensors/actuators.

Delivery of the exercise.

This exercise must be delivered at the campusvirtual inside the task enable for this purpose. The maximum day of the delivery is the 22nd of October of 2026 at the 23:55 h in the proper space for the delivery

Important notes for the delivery.

Your delivery needs to be named:

AME26_27_DEVELOPER1_DEVELOPER2_OPTIONAL_EXERCISE1.zip